

CPE-469 Distributed Systems

Programming Assignment 6

Replicated Banking Server (I)

In this programming assignment, we extend the previous programming assignment by building a replicated banking server. Each replicated server is its own docker container, and they communicate together using RPCs.

Create 3 replicas of the banking server, and for the purpose of this assignment only, assume that the first server is the assigned leader of the first term with no elections and that this server will not go down.

Recall, all client requests are routed to the leader. The leader appends the requests in its logfile with the current term and issues an append entries RPC on all followers. The followers add the log entry and acknowledge the leader.

Once the leader gets the majority of the ACKs of an entry in the current term, it can decide that this entry is committed. When an entry is committed the leader applies the changes to its database and updates its own committed index. The committed index is exchanged in future append entries RPC. Refer to figure 2 in the Raft paper for all the append entries RPC parameters.

Recall in the Raft paper, the read-only requests are not included in the logfile and require specific handling. However, for simplicity, treat all read-only requests the same as the write requests by appending them in the logfile, and responding after committing the entry.

To test your replicated server, have a separate Docker container called 'Tester', that receives a request to compare all logfiles of the 3 replicated servers.

You should submit Docker, go and README files.